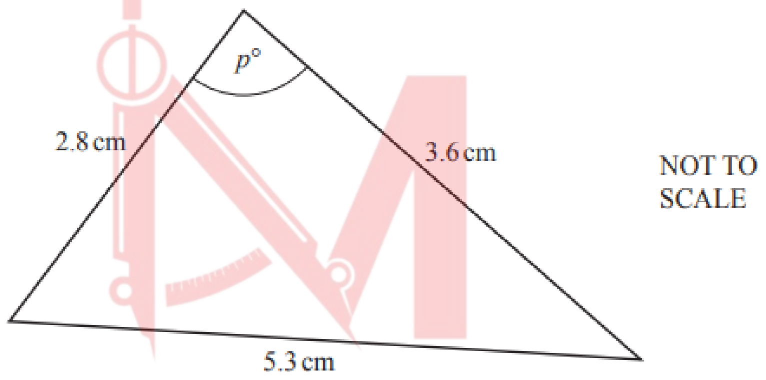
Trigonometry

Worksheet 2a

Further Trigonometry

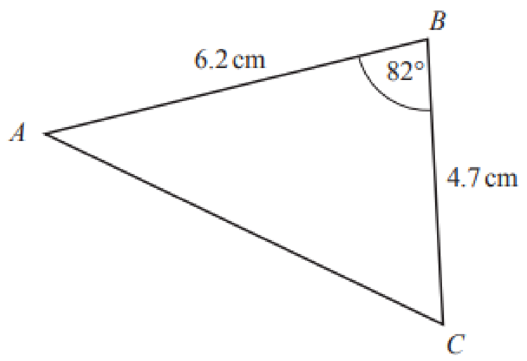
1. **M16/qp22/q15**



Find the value of p. [4]

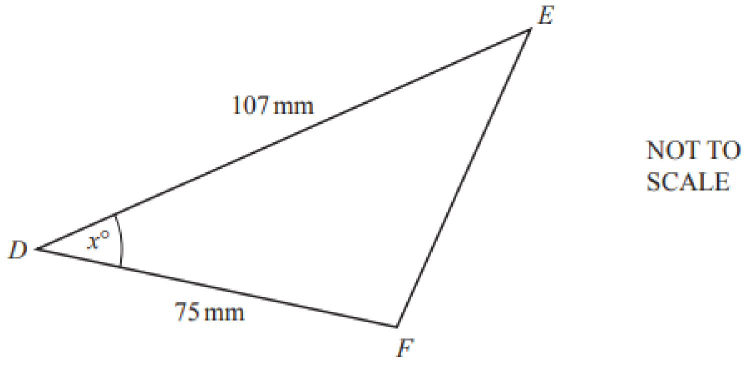
2. **W16/qp23/q21**

a.



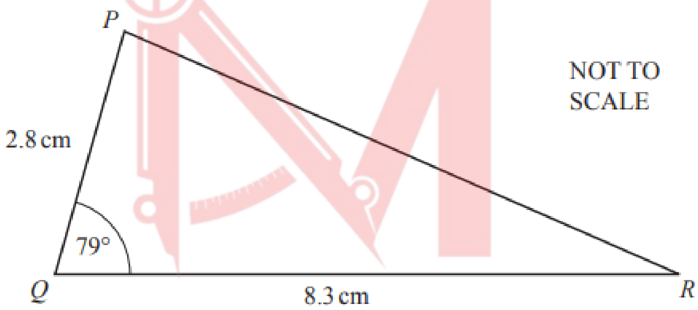
Calculate the area of triangle ABC. [2]

b.



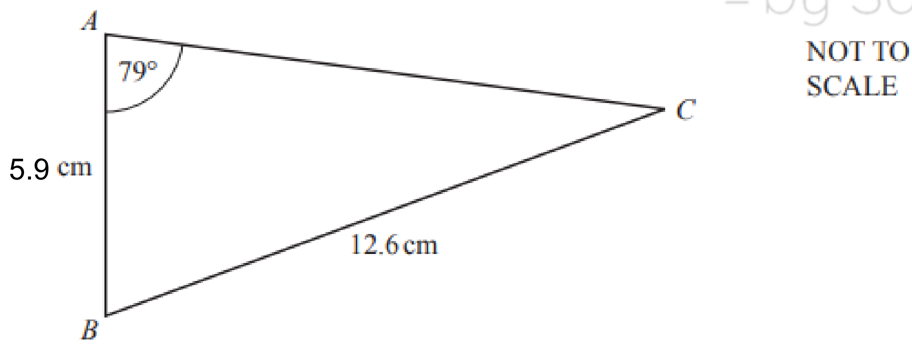
The area of triangle DEF is 2050 mm^2. Work out the value of x. [2]

3. M17/qp22/q19



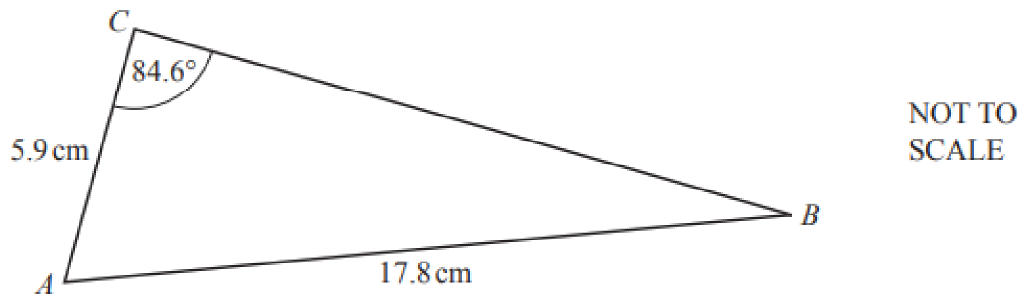
a. Calculate the area of triangle PQR. [2]

4. W17/qp23/q18



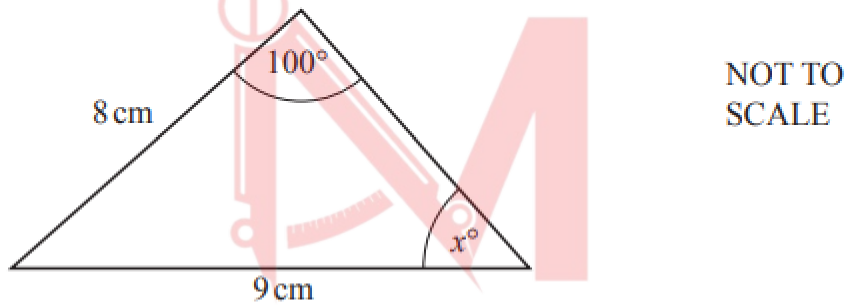
Calculate angle ABC. [4]

5. S18/qp21/q14



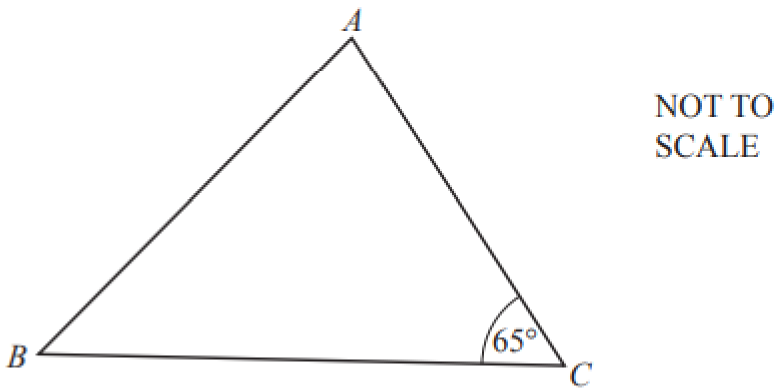
Use the sine rule to find angle ABC. [3]

6. W20/qp22/q19



a. Calculate the value of x. [3]
b. Calculate the area of the triangle. [3]

7. S21/qp23/q21



The shortest distance from B to AC is 12.8cm. Calculate BC. [3]

Answer Key

1. M16/qp22/q15

15	111.2 or 111.1 to 111.2	**4**	**M2** for [cos =] $\frac{2.8^2 + 3.6^2 - 5.3^2}{2 \times 2.8 \times 3.6}$ or **M1** for implicit form **A1** for [cos =] −0.362 to −0.361

2. W16/qp23/q21

21 (a)	14.4 or 14.42 to 14.43	**2**	**M1** for $\frac{1}{2} \times 6.2 \times 4.7 \times \sin 82$ oe
(b)	30.7 or 30.72…	**2**	**M1** for $\sin = \frac{2050}{\frac{1}{2} \times 107 \times 75}$

3. M17/qp22/q19

19 (a)	11.4 or 11.40 to 11.41	**2**	**M1** for $\frac{1}{2} \times 2.8 \times 8.3 \times \sin 79$ oe

4. W17/qp23/q18

18	73.6 or 73.63 to 73.64	**4**	**B3** for 27.4 or 27.36… OR **M2** for $\frac{5.9 \sin 79}{12.6}$ oe or **M1** for $\frac{\sin[C]}{5.9} = \frac{\sin 79}{12.6}$ oe and **M1dep** for 180 – 79 – *their C* (dep on at least **M1** earned)

5. S18/qp21/q14

14	19.3 or 19.26 to 19.27 nfww	**3**	**M2** for $[\sin =] 5.9 \times \frac{\sin 84.6}{17.8}$ or **M1** for $\frac{5.9}{\sin B} = \frac{17.8}{\sin 84.6}$ oe

6. W20/qp22/q19

19(a)	61.1 or 61.08 to 61.09...	3	**M2** for [$\sin x =$] $\frac{8\sin 100}{9}$ oe or better or **M1** for $\frac{9}{\sin 100} = \frac{8}{\sin x}$ oe
19(b)	11.7 or 11.66 to 11.67	3	**M2** for $\frac{1}{2} \times 9 \times 8 \times \sin(180 - 100 - \textit{their}\ \textbf{(a)})$ oe or **M1** for 180 – 100 – *their* **(a)**

7. S21/qp23/q21

21	14.1 or 14.12...	3	**M2** for $\sin 65 = \frac{12.8}{BC}$ oe or better or **M1** for recognition that the line from B is perpendicular to AC